

ECON 5345 Homework 9 Report

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1. The endogenous variables are

$$x_t = \begin{bmatrix} \hat{c}_t & \hat{i}_t & \hat{y}_t & \hat{l}_t \end{bmatrix}'.$$

The state variables are

$$s_t = \begin{bmatrix} \hat{k}_{t-1} & \hat{z}_t \end{bmatrix}'.$$

The linearized equilibrium conditions are

$$\begin{aligned} \sigma(\mathbb{E}_t \hat{c}_{t+1} - \hat{c}_t) &= \beta R^* \mathbb{E}_t(\hat{y}_{t+1} - \hat{k}_t) + \beta \phi \mathbb{E}_t(\hat{k}_{t+1} - \hat{k}_t) - \phi(\hat{k}_t - \hat{k}_{t-1}), \\ \left(1 + \frac{1}{\eta}\right) \hat{l}_t &= -\sigma \hat{c}_t + \hat{y}_t, \\ \hat{y}_t &= \hat{z}_t + \alpha \hat{k}_{t-1} + (1 - \alpha) \hat{l}_t, \\ \hat{y}_t &= \frac{C^*}{Y^*} \hat{c}_t + \frac{I^*}{Y^*} \hat{i}_t, \\ \hat{k}_t &= (1 - \delta) \hat{k}_{t-1} + \delta \hat{i}_t, \\ \hat{z}_t &= \rho \hat{z}_{t-1} + \varepsilon_t, \quad \varepsilon_t \sim N(0, \sigma_\varepsilon^2). \end{aligned}$$

For a given $\theta = (\rho, \phi, \sigma_\varepsilon)$, solving the linear rational expectations model yields the reduced-form decision rule

$$x_t = H(\theta) s_t,$$

where

$$H(\theta) = \begin{bmatrix} h_{ck}(\theta) & h_{cz}(\theta) \\ h_{ik}(\theta) & h_{iz}(\theta) \\ h_{yk}(\theta) & h_{yz}(\theta) \\ h_{lk}(\theta) & h_{lz}(\theta) \end{bmatrix}.$$

Then the capital accumulation equation becomes

$$\hat{k}_t = (1 - \delta)\hat{k}_{t-1} + \delta\hat{i}_t \quad (1)$$

$$= \left[(1 - \delta) + \delta h_{ik}(\theta) \right] \hat{k}_{t-1} + \delta h_{iz}(\theta) \hat{z}_t. \quad (2)$$

Thus the state transition equation is

$$s_{t+1} = \begin{bmatrix} \hat{k}_t \\ \hat{z}_{t+1} \end{bmatrix} = \underbrace{\begin{bmatrix} (1 - \delta) + \delta h_{ik}(\theta) & \delta h_{iz}(\theta) \\ 0 & \rho \end{bmatrix}}_{B(\theta)} s_t + \underbrace{\begin{bmatrix} 0 \\ \sigma_\varepsilon \end{bmatrix}}_{u_{t+1}} w_{t+1}. \quad (3)$$

where $w_{t+1} \sim N(0, 1)$ is the system error.

The observed data are measured with error:

$$x_t^* = x_t + e_t, \quad e_t \sim N(0, \Xi), \quad \Xi = \text{diag}(0.25, 0.25, 0.25, 0.25). \quad (4)$$

Therefore the measurement equation is

$$x_t^* = H(\theta)s_t + e_t. \quad (5)$$

Hence the state-space representation is

$$s_{t+1} = B(\theta)s_t + u_{t+1}, \quad (6)$$

$$x_t^* = H(\theta)s_t + e_t. \quad (7)$$

We do not need to recompute the steady state for each candidate θ because ρ and σ_ε affect only the shock process, while ϕ affects only transition dynamics and drops out of the steady-state conditions. Therefore the steady state depends only on the known parameters and remains unchanged across candidate values of θ .

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